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VPC Monitoring with Flow Logs

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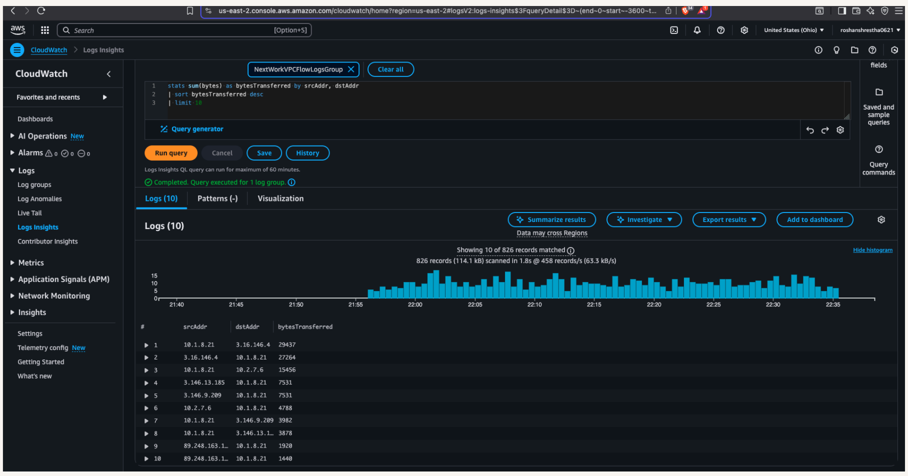
Roshan Shrestha



Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a logically isolated network in the amazon cloudspace. It is useful because it resembles a traditional network that we would operate in our own data center.

How I used Amazon VPC in this project

In today's project, I used Amazon VPC to create two EC2 instances, one in each VPC. Open a peering connection between the two VPCs. Set up Flow logs using CloudWatch. Test ping messages between the two instances and analyze the traffic using flow log

One thing I didn't expect in this project was...

One thing I didn't expect in this project was being able to query into the flow logs.

This project took me...

This project took me about 2 hrs.

In the first part of my project...

Step 1 - Set up VPCs

In this step, we are creating two VPCs from scratch which we will use to setup a perring network connection between them.

Step 2 - Launch EC2 instances

In this step, we are launching an EC2 instance in each VPC's public subnet to test a peering connection.

Step 3 - Set up Logs

In this step, we are setting up a way to track all inbound and outbound network traffic and also setting up a space that stores all of these records.

Step 4 - Set IAM permissions for Logs

In this step, we are going to give VPC Flow Logs the permission to write logs and send them to CloudWatch.

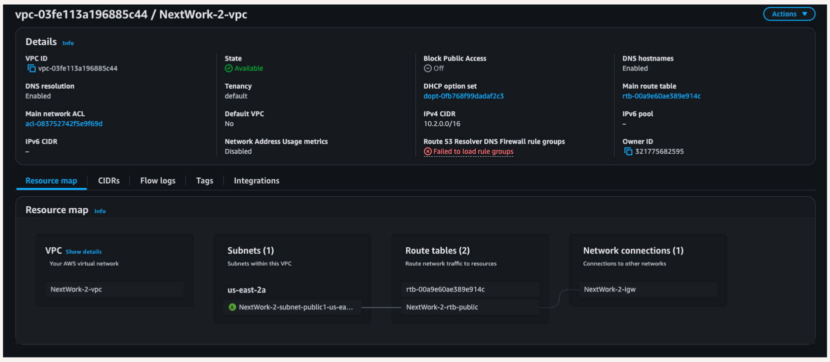
Multi-VPC Architecture

I started my project by launching two VPCs. I created two public subnets - one for each VPC.

The CIDR blocks for VPCs 1 and 2 are 10.1.0.0/16 and 10.2.0.0/16. They have to be unique because we don't want any overlapping IP addresses which result in routing problems and connectivity issues.

I also launched EC2 instances in each subnet

My EC2 instances security groups allow all inbound SSH and ICMP - IPv4 traffic from all sources and allow all outbound traffic. This is because we need the ICMP - IPv4 to be able to ping from one instance to another and to the public internet.



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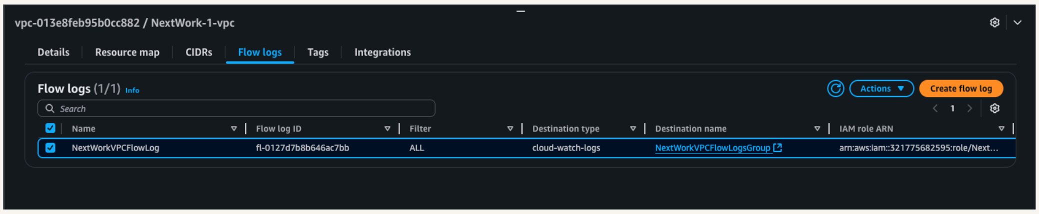
Logs

Logs are records of everything happening with our VPCs.

Log groups are a big folder in AWS where you keep related logs together. They are region specific.

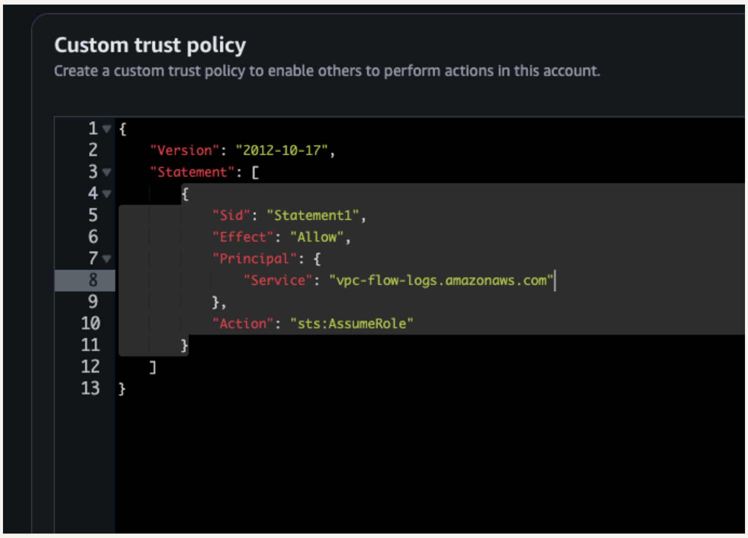


IAM Policy and Roles

I created an IAM policy because VPC Flow Logs by default don't have the permission to record logs and store them in your CloudWatch log group. This policy makes sure that your VPC can now send log data to your log group.

I also created an IAM role because we cannot assign IAM policies directly to resources. Policies are combined under a role and that role is assigned to a resource.

A custom trust policy is a specific type of policy, different from IAM policies. While IAM policies help you define the actions a user/service can or cannot do, custom trust policies are used to very narrowly define who can use a role.


Custom trust policy
Create a custom trust policy to enable others to perform actions in this account.
{
"Version": "2012-10-17",
"Statement": [
{
"Sid": "Statement1",
"Effect": "Allow",
"Principal": {
"Service": "vpc-flow-logs.amazonaws.com"
},
"Action": "sts:AssumeRole"
}
]
}

In the second part of my project...

Step 5 - Ping testing and troubleshooting

In this step we are going to setup VPC peering and send test messages from EC2 instance 1 to 2 because we want to generate some network traffic and see whether our flow logs can pick up on them.

Step 6 - Set up a peering connection

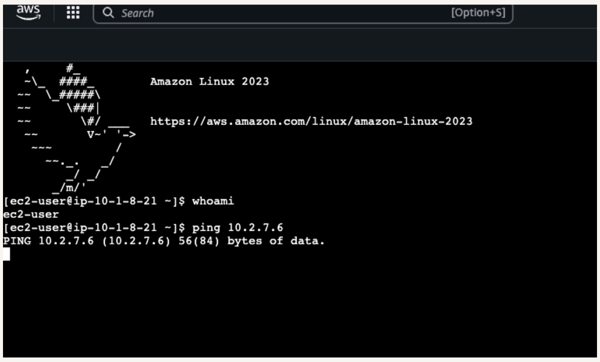
In this step we are creating a peering connection to crete a direct route between VPCs 1 and 2.

Step 7 - Analyze flow logs

In this step we are reviewing the flow logs recorded about VPC 1's public subnet and analyzing the logs.

Connectivity troubleshooting

My first ping test between my EC2 instances had no replies, which means the two instances are not able to communicate with each other. The request from instance 1 is not reaching instance 2.



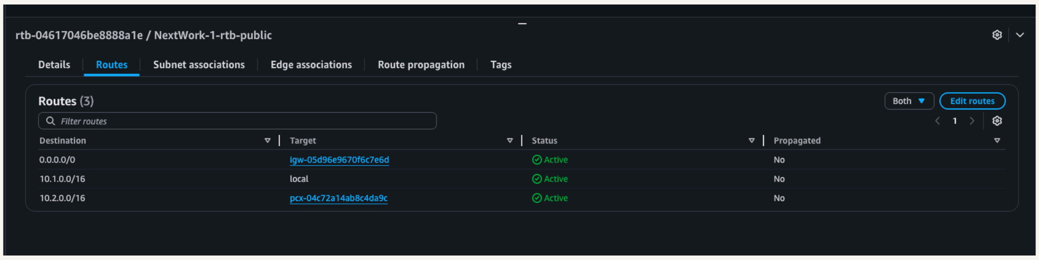
I could receive ping replies if I ran the ping test using the other instance's public IP address, which means the two instances are able to communicate using the public IP which uses the public internet.

Connectivity troubleshooting

Looking at VPC 1's route table, I identified that the ping test with Instance 2's private address failed because we don't have a direct route to VPC2 from VPC 1.

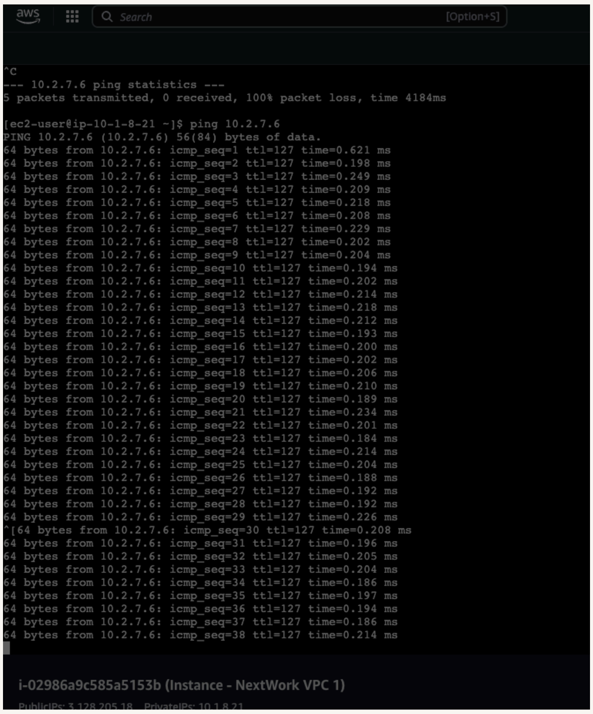
To solve this, I set up a peering connection between my VPCs

I also updated both VPCs' route tables so that both VPCs can directly communicate to each other using their private addresses utilizing the peering connection VPC 1 <> VPC 2.



Connectivity troubleshooting

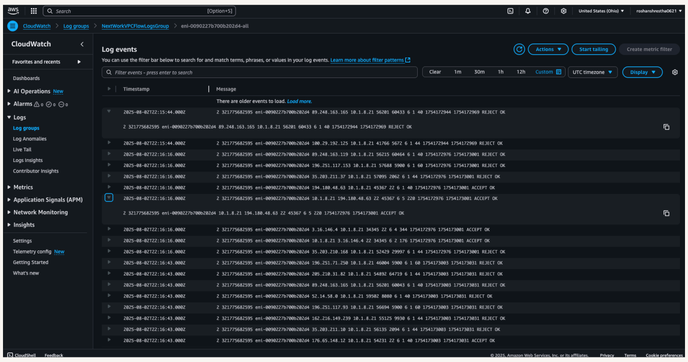
I received ping replies from Instance 2's private IP address! This means that the two instances are successfully using the peering connection we setup VPC 1 <> VPC 2.



Analyzing flow logs

Flow logs tell us about the version of the flow log, the AWS account that creatded it, network interface, source and destination IPs and ports, no. of data packets and bytes, start/end time, if the traffic was allowed in and if looging was successful

The flow log I've captured tells us about the different sources that pinged the EC2 instance 2 and if the ping was successful.



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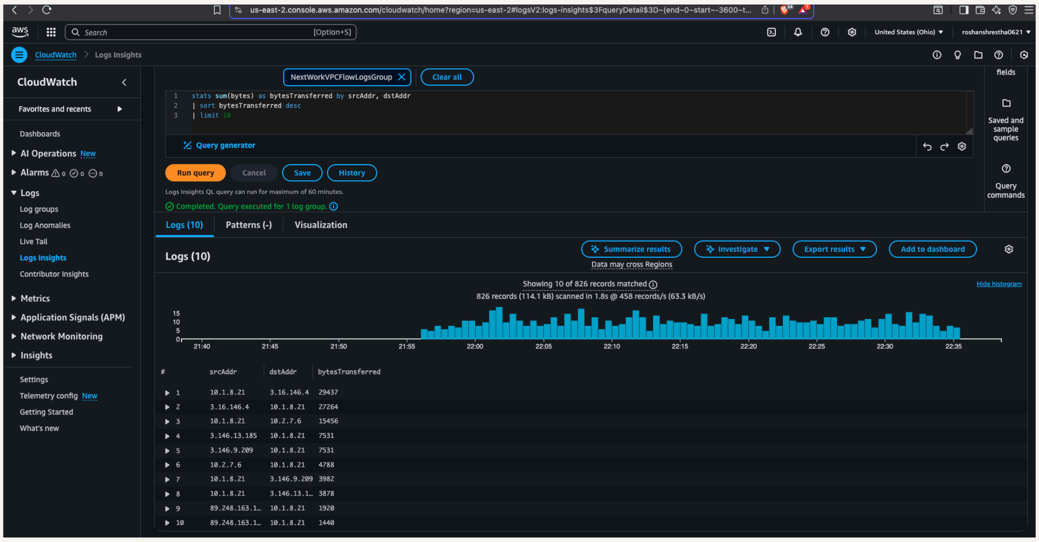
Logs Insights

Logs Insights is a CloudWatch feature that analyzes your logs. In Log Insights, you use queries to filter, process and combine data to help you troubleshoot problems or better understand your network traffic.

I ran the query "Top 10 byte transfers by source and destination IP addresses". This query analyzes the most data transferred between source and destination IP addresses.



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